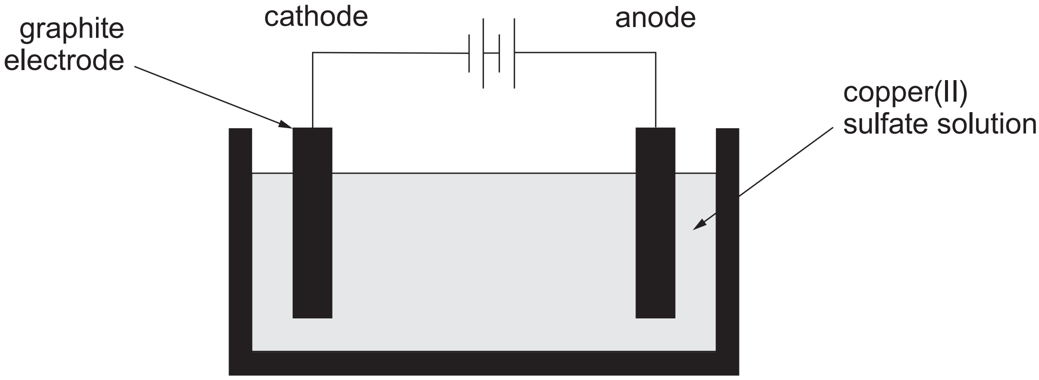
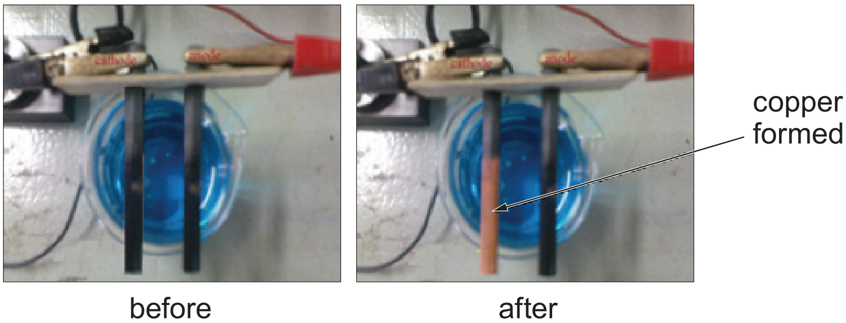
7. A group of students carried out an investigation into the electrolysis of copper(II) sulfate solution. They used the apparatus shown to test the hypothesis:

"the mass of copper that forms on the cathode increases as the time increases"





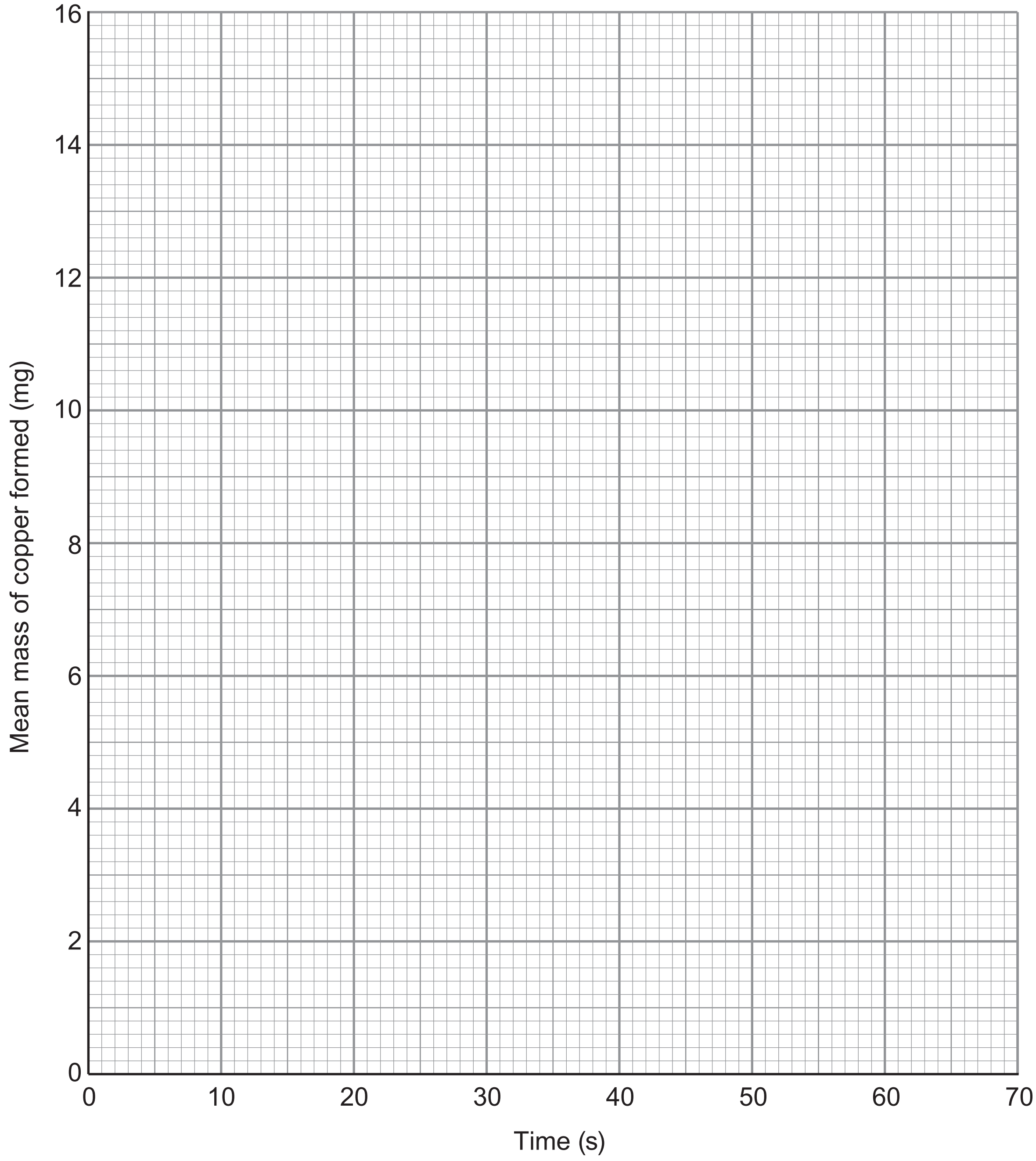
To test the hypothesis, they weighed the cathode before placing it into the copper(II) sulfate solution and then again after allowing electrolysis to take place for varying times.

Their results are shown below.

Time (s)	Mass of copper formed (mg)		
	1	2	Mean
0	0	0	0
10	2.8	3.2	3.0
20	4.8	5.0	4.9
30	8.2	7.8	8.0
40	10.8	11.2	11.0
50	12.9	13.1	13.0
60	15.8	16.0	15.9

Examiner only

(a) On the grid below, plot the mean mass of copper formed against time. Draw a suitable line. [3]



Examiner only

(b) (i) Use the results collected at 30 s and the following equation to calculate the percentage variation in these measurements. [2]

$$\text{percentage variation} = \frac{\text{furthest mass from the mean} - \text{mean mass}}{\text{mean mass}} \times 100$$

Percentage variation = ... %

(ii) The mass of copper formed is lower than expected. Give the most likely reason for this difference. [1]

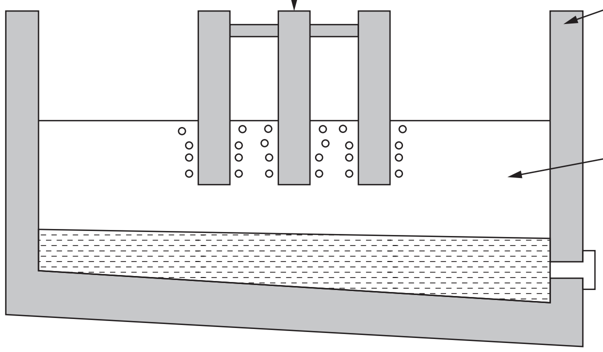
....

....

....

Examiner only

(c) (i) Aluminium is extracted from molten aluminium oxide by electrolysis.



I. Explain why aluminium forms at the cathode. [2]

....

....

....

II. Complete and balance the equation for the overall reaction that takes place. [2]

$2Al_2O_3 \longrightarrow \square$ .... + $\square$ ....

(ii) Potassium can also be extracted through electrolysis of potassium carbonate.

Write the **formula** of potassium carbonate to complete the equation for the overall reaction. [1]

2 .... $\longrightarrow 4K + 2CO_2 + O_2$

11

Turn over.